



PROJECT SYNOPSIS

Submitted for Review I

Smart Classroom Infrastructure Monitoring and Maintenance System Using Computer Vision and IoT

Degree / Programme	B.Tech.
Department	CSE (AI & ML) and E&TC
Academic Year	2026 - 2027
Project Guide	Dr. P. R. Sonawane
Date of Submission	19/08/2026

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1. Abstract

Schools and colleges run large numbers of classrooms and laboratories, and the furniture, lighting, boards and fans inside them need regular upkeep. Most institutions still catch problems through informal or periodic manual inspection, which leaves faults undetected for days, produces no reliable record of what was reported, and gives no way to confirm whether a reported issue was actually fixed. This project addresses that gap with a Smart Classroom Infrastructure Monitoring and Maintenance System built on computer vision and IoT sensing. A fixed camera and an ESP32 sensor node observe each classroom. Vision handles occupancy, board condition and fan operation; the sensor node measures electrical and environmental parameters directly. Both streams are relayed to a central server over MQTT, where observations are validated before anything is logged as a maintenance issue — a single abnormal reading never raises a ticket on its own. The system instead waits for repeated, corroborating evidence across sources before treating a condition as a genuine fault, which is what keeps false alerts down.

Once a fault is confirmed, it is automatically classified, prioritised and turned into a maintenance ticket routed to the right staff member. Supervisors and maintenance staff track that ticket through a web application from initial report to verified resolution, so the record runs unbroken from detection to closure rather than stopping at a dashboard reading. What makes the system worth building is not any one detection capability on its own, but combining vision and sensor evidence into a single decision — that combination is what lets it tell ordinary classroom activity apart from a condition that actually needs attention. The result is a system aimed at fewer false alerts, a clearer accountability trail, and infrastructure that gets managed rather than merely observed.

Keywords: Internet of Things, Computer Vision, Edge Computing, MQTT, Smart Classroom, Predictive Maintenance, Energy Monitoring.

2. Introduction

2.1 Background

Educational institutions such as engineering colleges have a large number of classrooms, laboratories, and seminar halls that are used regularly by students and faculty. These spaces contain important facilities such as lights, fans, electrical equipment, boards, and furniture. Keeping these facilities in proper working condition is important for providing a comfortable and effective learning environment. However, their maintenance is often dependent on periodic inspection and manual reporting by staff and faculty. As the number of classrooms increases, it becomes difficult to monitor every room regularly. Because of this, problems such as faulty electrical equipment, unnecessary power consumption, unclean boards, or damaged furniture may remain unnoticed for a long time. In addition, manually reported problems may not always have a proper digital record, making it difficult to track whether the issue has been assigned, repaired, and properly resolved.

The use of Computer Vision and Internet of Things (IoT) technologies can help improve this process. Computer vision has been used for classroom attendance and attention monitoring [1], faculty presence detection using multiple sensing sources [2], classroom occupancy detection [4], and classroom behaviour recognition [5]. Similarly, IoT-based systems have been used for classroom automation using occupancy and environmental sensors [3] and for occupancy monitoring in smart buildings to improve energy efficiency [6]. These studies show that vision and sensor technologies can be useful for monitoring different aspects of classroom and building environments. However, most existing systems focus on a particular task such as attendance, presence, occupancy, behaviour recognition, or automatic control. They generally do not focus on continuously monitoring classroom infrastructure and connecting detected problems with an actual maintenance process.

Another important issue is that a single observation may not always be enough to decide whether a fault has occurred. For example, detecting electrical power consumption alone cannot confirm energy wastage because the classroom may be occupied. Similarly, detecting an empty classroom does not indicate whether any electrical equipment is unnecessarily running. Combining information from both sources can provide better context and make the fault decision more reliable. Multimodal sensing with confirmation mechanisms has also shown the potential to reduce false alarms in other application areas [7]. However, this type of confirmation is not sufficiently explored for classroom infrastructure monitoring. Therefore, there is a need for a system that combines computer vision and IoT sensing to monitor classroom conditions, verify abnormal situations, and generate maintenance actions only when a fault is confirmed. The proposed Smart Classroom Infrastructure Monitoring and Maintenance System (SIMMS) addresses this need by combining monitoring, fault verification, and digital maintenance management into a single workflow, allowing an issue to be followed from detection and ticket generation to assignment, repair, and verified closure.

2.2 Problem/Research Gap

Existing classroom monitoring systems mainly focus on individual tasks such as attendance, faculty presence, occupancy, classroom automation, and student behaviour using computer vision or IoT [1]–[6]. Although these studies demonstrate the usefulness of these technologies, they generally do not combine visual and sensor information to verify classroom infrastructure faults.

Another limitation is that monitoring usually ends at detection or alert generation. There is limited integration between fault detection and maintenance management, such as automatic ticket creation, prioritisation, assignment, and verified closure. In addition, temporary abnormal readings may generate false alerts if they are not confirmed over multiple observations. Multimodal confirmation has shown potential for reducing false alarms in other domains [7], but this approach is not sufficiently explored for classroom infrastructure monitoring.

Therefore, there is a need for a unified system that combines computer vision and IoT sensing, verifies abnormalities using contextual and temporal information, and connects confirmed faults to an end-to-end digital maintenance workflow. The proposed system addresses these gaps through multimodal fault verification, false-alert reduction, and automated maintenance ticket management.

2.3 Original Contributions

The original contributions of the proposed work can be summarized as follows:

- **Integration of computer vision and IoT:** Development of a unified monitoring system that combines visual observations with electrical and sensor-based information for classroom infrastructure monitoring.
- **Context-based fault verification:** Use of information from multiple sources and repeated observations to distinguish persistent infrastructure problems from temporary or normal classroom conditions.
- **Automated maintenance management:** Conversion of confirmed abnormalities into classified and prioritized maintenance tickets that can be assigned to the appropriate staff.
- **End-to-end maintenance tracking:** Development of a digital workflow that tracks an issue from detection and reporting through repair, verification, and closure.
- **Reduction of false maintenance alerts:** Application of temporal confirmation, issue de-duplication, and cooldown mechanisms to improve the reliability of automated ticket generation.

3. Literature Survey and Research Gap

The reviewed literature is grouped into four themes: vision-based classroom monitoring, IoT-based classroom and building monitoring, multimodal sensing with confirmation, and the enabling technologies and maintenance-management practices on which the proposed system depends. Each study is summarised below and compared in Table 1.

3.1 Review of Existing Work

A. Vision-based classroom monitoring

Muzayanah et al. [1] apply an OpenCV face-detection pipeline to mark attendance and to raise an alert once a face has been absent from consecutive frames for a fixed duration, feeding the result into a classroom management dashboard. The design establishes that one fixed camera at the front of a room is adequate for routine observation, and that a simple absence timer suppresses momentary detection dropouts. The pipeline is nevertheless vision-only and person-centred: a detection is never

corroborated by an independent sensor, occupancy is the end product rather than context, and the output terminates at a report rather than a maintenance action.

Monti et al. [4] fine-tune a pretrained convolutional backbone by transfer learning and deploy it on a Raspberry Pi for people counting across several classroom layouts in a smart-campus installation, avoiding both large-scale dataset collection and server-side GPU inference. The work is directly relevant here as evidence that ARM-class CPU inference suffices when the sampling rate is low and the task is coarse counting rather than tracking. Occupancy remains the final output, however: the counts are never related to electrical load or to the physical condition of the room, and no maintenance action follows from them.

Wang et al. and Chen et al. [5] add attention mechanisms to YOLO detectors for classroom behaviour recognition - a coordinate attention module in CA-YOLOv7 and an improved YOLOv8 variant - targeting small, densely seated and partially occluded students, and report performance as mean average precision at an IoU threshold of 0.5. Their results establish YOLO-family detectors as dependable on classroom scenes, including at the small object scales that seat-level analysis requires, which supports the choice of backbone in this project. The objective is behavioural analytics of people rather than the state of the room and its equipment.

B. IoT-based classroom and building monitoring

Niveditha et al. [3] combine PIR motion sensing, an LDR for ambient light and a temperature sensor to actuate relays controlling classroom lights and fans, cutting consumption in unoccupied rooms and confirming that microcontroller-class hardware is adequate for classroom energy control. Actuation is driven by fixed thresholds applied to each sensor independently, with no hysteresis or temporal voting across successive readings. Without visual context the system cannot separate a genuinely vacant room from one holding a stationary occupant - the classic PIR failure mode - and it keeps no record of equipment faults.

Zhai et al. [6] use occupancy information to drive HVAC and lighting in a demand-controlled rather than schedule-driven regime, quantifying the energy available from occupancy-aware operation and providing the rationale for treating loads left active in a vacant room as waste. The sensing is indirect and single-modality: the system can infer that a space is unoccupied, but it cannot verify that the equipment it switches is actually functioning, and it carries no notion of a fault, a ticket or a verified closure.

Candanedo and Feldheim [8] classify light, temperature, humidity, carbon dioxide and derived humidity-ratio measurements from an office room using linear discriminant analysis, classification trees and random forests, with ground truth established from time-stamped photographs; reported accuracy falls in the region of 95 to 99 % depending on the model and the feature subset, with illuminance the single most informative predictor. The study is the standard reference for how far environmental signals alone can be taken without a camera. Performance is tied to the ventilation dynamics of the specific room, and the output is a binary room state - it identifies neither which seats are occupied nor whether any equipment is faulty.

Chen, Jiang and Xie [9] review camera, PIR, carbon dioxide, Wi-Fi and hybrid approaches to occupancy sensing and separate the problem by occupancy resolution - presence, count, location and identity - noting that accuracy requirements and error behaviour differ sharply at each level. Their central finding, that hybrid multi-sensor schemes consistently outperform any single modality, is the methodological justification for a fused design. The review also observes that most surveyed systems terminate at an estimate with no downstream action defined, which is precisely the gap this project targets.

C. Multimodal sensing and false-alarm confirmation

Kinjarapu et al. [2] infer faculty presence on a Raspberry Pi by combining PIR motion, face detection, voice activity and connected-device network signals, so that the decision is taken at the fusion stage rather than by any single modality. This is the closest existing work to the principle adopted here and demonstrates that decision-level fusion of heterogeneous signals is practical on constrained edge hardware. Its scope ends at establishing presence: no fault condition is defined, no confirmation window or debounce behaviour is reported, and no maintenance workflow follows.

Kumar et al. [7] fuse PIR, ultrasonic ranging and camera data with probabilistic confirmation, raising an obstruction alert only when independent sources agree, explicitly in order to suppress the false positives that would otherwise render the system unusable in a safety setting. Although the domain is railway safety, this is the clearest published demonstration of the principle on which the present work depends - cross-modal agreement as the mechanism that makes automated alerts trustworthy. The logic terminates in an instantaneous alert rather than in a maintenance task with a lifecycle, an assignee and a verified closure.

D. Enabling technologies and maintenance management

Shi et al. [10] argue that the volume of data generated at the network edge makes cloud-only processing untenable on latency, bandwidth and privacy grounds, and reframe the edge device as both a consumer and a producer of data rather than a passive sensor. The paper is the architectural justification for analysing frames on a Raspberry Pi and publishing only derived JSON observations, so that raw video never leaves the classroom. It is a position and roadmap paper rather than an implementation, and offers no deployment specifics for constrained vision workloads.

Wang, Bochkovski and Liao [11] introduce YOLOv7, combining an extended efficient layer aggregation network (E-ELAN), planned re-parameterised convolution and coarse-to-fine auxiliary head label assignment - a trainable bag-of-freebies that raises accuracy without increasing inference cost - to reach 56.8 % average precision on MS COCO across an operating range of roughly 5 to 160 frames per second. This underpins the treatment of person detection here as a mature, reusable capability rather than a component requiring training. The reported figures assume GPU inference, so a nano-scale variant is used in this project to fit the Raspberry Pi budget at a 30 to 60 second sampling interval.

Cha, Choi and Buyukozturk [12] train a convolutional network on the order of 40,000 image patches of 256 x 256 pixels to detect concrete cracks directly from imagery, reporting approximately 98 % classification accuracy and clear robustness advantages over Canny and Sobel edge-detection baselines under varying illumination and shadow. The work is widely cited as the reference point for vision-based infrastructure defect detection and shows that defect classification is learnable from images alone. It also depends on a large annotated defect dataset, which is the reason visible furniture damage is scoped here as a proof of concept rather than a committed deliverable.

Carvalho et al. [13] survey machine-learning methods applied to predictive maintenance, finding random forests, artificial neural networks and support vector machines to be the most frequently applied algorithms, and the reported work to be concentrated on rotating machinery for which labelled failure histories already exist. The review positions maintenance prediction as a data problem resting on a reliable fault record rather than an algorithmic one. It notes that such records are seldom available in practice, which motivates the structured audit trail generated here as a by-product of ticket closure.

3.2 Comparative Analysis

Table 1: Comparative Analysis of Existing Classroom Monitoring Systems and Enabling Work

No.	Study	Domain	Modality	Method	Limitation
1	Muzayanah et al. (2024) ¹	Classroom Monitoring	Vision	Face detection with OpenCV; alerts when a student is absent for a fixed time	Vision only; no sensor fusion or maintenance monitoring
2	Kinjarapu et al. (2026) ²	Faculty presence	Multimodal	Combines PIR, face, voice, and device signals on Raspberry Pi	Focuses only on presence; no fault detection or maintenance workflow
3	Niveditha et al. (2023) ³	Classroom Automation	IOT	Uses PIR, LDR, and temperature sensors to control lights and fans	No vision; uses simple thresholds; no fault or maintenance tracking
4	Monti et al. (2022) ⁴	Smart campus occupancy	Vision	Raspberry Pi-based people counting across different classroom layouts	Only counts people; no sensor fusion or infrastructure monitoring
5	Wang et al.; Chen et al. (2024) ⁵	Teaching Behaviour	Vision	YOLO-based recognition of classroom activities	IoT mainly supports data transfer; not used for decision fusion or maintenance
6	IEEE occupancy survey (2015) ⁶	Smart Buildings	IOT	Occupancy-based control of HVAC and lighting	No vision, fault detection, ticketing, or closure verification
7	Kumar et al., NETRA (2026) ⁷	Railway obstruction detection	Multimodal	Combines PIR, ultrasonic, and camera data with probabilistic confirmation	Different application domain, but supports the use of sensor confirmation to reduce false alarms
8	Candanedo and Feldheim (2016) ⁸	Office occupancy	IoT	LDA, classification trees and random forest on light, temperature, humidity and CO2	No vision; binary room state only; no seat-level detail or fault detection
9	Chen, Jiang and Xie (2018) ⁹	Building occupancy (review)	Multimodal	Reviews camera, PIR, CO2, Wi-Fi and hybrid sensing by occupancy resolution	Survey only; reviewed systems end at an occupancy estimate with no action defined
10	Shi et al. (2016) ¹⁰	Edge computing	Concept	Case for processing at the source on latency, bandwidth and privacy grounds	Position paper; no implementation or guidance for constrained vision workloads
11	Wang, Bochkovski and Liao (2023) ¹¹	Object detection	Vision	YOLOv7: E-ELAN, re-parameterised convolution, auxiliary-head label assignment	GPU-benchmarked; generic detector, not applied to infrastructure or maintenance

No.	Study	Domain	Modality	Method	Limitation
12	Cha, Choi and Buyukozturk (2017) ¹²	Infrastructure defect detection	Vision	CNN crack detection from image patches, compared against Canny and Sobel baselines	Requires a large annotated defect dataset; ends at detection
13	Carvalho et al. (2019) ¹³	Predictive maintenance (review)	Data-driven	Review of ML methods; random forest, ANN and SVM most frequently applied	Assumes an existing labelled failure history; no sensing or ticketing workflow

3.3 Identified Research Gap

1. Modalities are not fused for decision-making. Vision systems and IoT systems are typically reported separately. Where both appear, sensor data is presented alongside vision output rather than combined into a single rule. A fault such as power being consumed while the room is confirmed empty requires both signals and is therefore rarely addressed.
2. Detection is not connected to resolution. Automated detection work ends at an accuracy figure, while maintenance-management work begins at manual data entry. Very little work spans both, so no system demonstrates a fault progressing automatically from detection to verified closure.
3. False-alarm behaviour is not reported. Published systems report accuracy, precision, recall and mAP, all measured on curated test sets. Almost none report how many spurious alerts the system generates during continuous unattended operation, which is the metric that determines whether maintenance staff continue to use it.

This project addresses all three gaps: it fuses vision and sensor signals in every decision rule, spans the full detection-to-closure workflow, and treats operational false-alarm rate as a primary reported result.

4. Problem Statement

To design and develop an IoT based Smart Classroom Infrastructure Monitoring and Maintenance System by utilizing computer vision and sensor based data to automatically track and monitor the occupancy, board condition and operational status of light and fan inside the classroom. The system needs to check detected abnormalities with information from multiple sources and create maintenance ticket only if it is a confirmed abnormality. These tickets should be categorized, prioritized and then assigned to the correct maintenance staff and then monitored until they are closed by the supervisor. The intended system will address the issues of false maintenance alerts, accountability and a continuous digital workflow from fault detection to the verified resolution.

5. Aim and Objectives

5.1 Aim

Create a Smart Classroom Infrastructure Monitoring and Maintenance System that uses computer vision with edge devices and sensors to identify and alert to classroom infractions and electrical problems, confirm the detected defects and close and maintain infrastructure and electricals through a digital maintenance workflow from automated ticket generation to supervisor verified closure.

5.2 Core Objectives - Committed Deliverables

1. Periodically capture and analyse classroom condition on an edge computing device
2. Determine seat-level occupancy using person detection over calibrated seat regions
3. Monitor light and fan status and electrical parameters using IoT sensors
4. Confirm, de-duplicate and debounce detections before raising a ticket
5. Automatically classify, prioritise and assign maintenance tickets

6. Proposed System and Methodology

6.1 System Architecture

Table 3: System Architecture

Layer	Components	Function
Classroom - vision	Fixed camera (1 frame per 30-60 s); Raspberry Pi 4 edge device	Person detection, seat-ROI occupancy, board ink-ratio analysis, blurred evidence capture; publishes JSON observations only
Classroom - sensing	ESP32 node with PZEM-004T, SCT-013 CT clamp, DHT11 and LDR (sample per 5 s)	Voltage, current, power, power factor and energy; light and fan branch status; publishes JSON telemetry
Transport	MQTT via Eclipse Mosquitto	Carries observations and telemetry from classroom to server
Server	FastAPI backend with PostgreSQL 16	Ingestion and persistence, confirmation and de-duplication, rule engine and classification, ticket lifecycle and assignment, health score, REST API, Web Push and Telegram
Client	React Progressive Web Application	Supervisor: dashboard, verification and closure. Staff: assigned tickets and status updates. Admin: classroom calibration

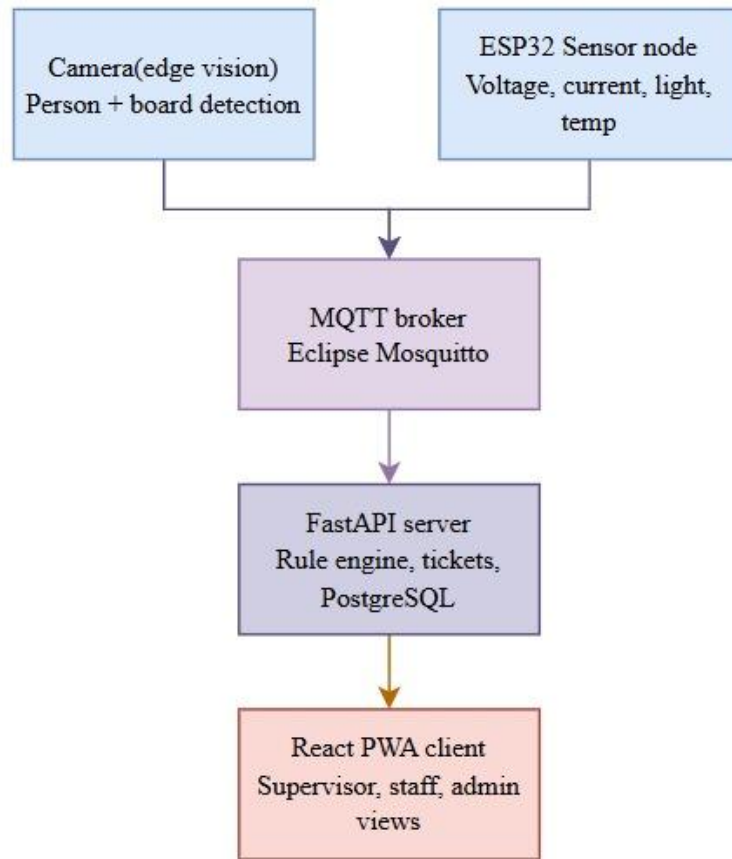


Fig. 1: System Architecture

Key design decision - periodic video inspection, not real time video. Classroom infrastructure evolves in minutes and hours, not milliseconds. By analysing one frame every 30-60 seconds, the Raspberry Pi 4 duty cycle can be kept below 7% and the position on privacy is defensible as there is no continuous video stream anywhere in the system.

Why MQTT over HTTP polling. The nodes in the classroom are low-power, with intermittent connections to the internet, so the choice for a publish-subscribe protocol was made over periodic HTTP requests, due to the QoS levels and persistent sessions in the publish-subscribe protocol to help minimize the loss of telemetry when the node loses internet connection, and the reduction in request overhead from polling every few seconds. This was taken into account when resorting to HTTP polling; it was rejected.

Why YOLO-nano, instead of a custom-trained detector? Detecting persons is a solved and general purpose problem and a large annotated dataset suitable for training a classroom-specific detector is not something that the project can practically gather. This avoids having to add classroom-specific reasoning to the machine learning detector itself and keeps the machine learning risk in the project to a minimum by using a pretrained YOLO-nano (person class only).

6.2 Classroom Calibration

Each classroom is configured once by admin through an HTML canvas, which is stacked on a reference image of the empty room, where the admin draws polygons for the seats, the 4 corners of the board and the regions for the fans. Calibrations are versioned to allow for historical data to be understood if furniture is moved. This human setup is made on purpose, and represents a trade of just a few minutes per room for a lot of machine-learning risk.

6.3 Detection Modules

Table 4: Detection Modules and Methods

Module	Method	Rationale
Occupancy	Pretrained YOLO (person class only), then test each calibrated seat ROI for overlap	Seat geometry is already known from calibration, so no seat detector needs training. Eliminates the project's largest annotation workload.
Board condition	Perspective-warp the board region, adaptive threshold, morphological opening, compute ink-pixel ratio	The task is a well-posed measurement, not a perception problem. Deterministic, explainable, tunable on site without retraining.
Electrical status	PZEM-004T on the demonstration panel (V, I, P, PF, energy over UART) plus one CT clamp on the fan branch	Expected current bands are learned during commissioning rather than hard-coded, so the system adapts to the loads actually installed.
Energy wastage	Occupancy = 0 for at least 10 continuous minutes AND measured load above threshold	Requires both modalities; neither sensor alone can distinguish waste from legitimate use.
Fan rotation (E1)	Inter-frame difference within the calibrated fan ROI; a rotating fan produces a high motion score	Cross-checks current against actual mechanical motion, diagnosing a seized motor that neither signal identifies alone.

Contextual coupling. The board module consumes the occupancy module's output: a board covered in writing during an active lecture is normal and raises nothing, whereas the same board with the room confirmed vacant for ten minutes generates a cleaning task. This inter-module dependency is the concrete form of the multimodal argument.

6.4 Event Confirmation, De-duplication and Cooldown

A raw detection is not an event, and an event is not a ticket. Three mechanisms are applied in sequence, and together they constitute the system's principal reliability contribution.

(a) Temporal confirmation. Each detector emits an observation carrying a state and a confidence. A sliding window of the last M observations per (classroom, detector, target) is maintained.

Table 5: Event Confirmation Parameters

Parameter	Value
Sampling interval	60 s (vision), 5 s (sensors)
Window size M	5 observations
Fault confirmation threshold N	4 of 5 must agree
Recovery threshold	3 of 5 clear
Minimum confidence to vote	0.55

Thresholds are deliberately asymmetric: the system is slower to raise a fault than to clear one, biasing it against false tickets.

(b) De-duplication by signature. All confirmed faults can be reduced to the following: classroom_id, issue_type, target_id, calibration_version. If the ticket is already open with that signature, then a new ticket is not opened, the Recurrence counter is increased, the observation is added to the Evidence trail and the priority escalates if it is exceeded.

(c) Cool down and auto-resolution. A signature is suppressed for 30 minutes after the closing of the ticket, so a ticket will not reopen during a technician's time on the job. If a fault resolves while the ticket is still not acted upon, the ticket will be automatically closed with a status of AUTO_RESOLVED.

6.5 Ticket Lifecycle, Classification and Priority

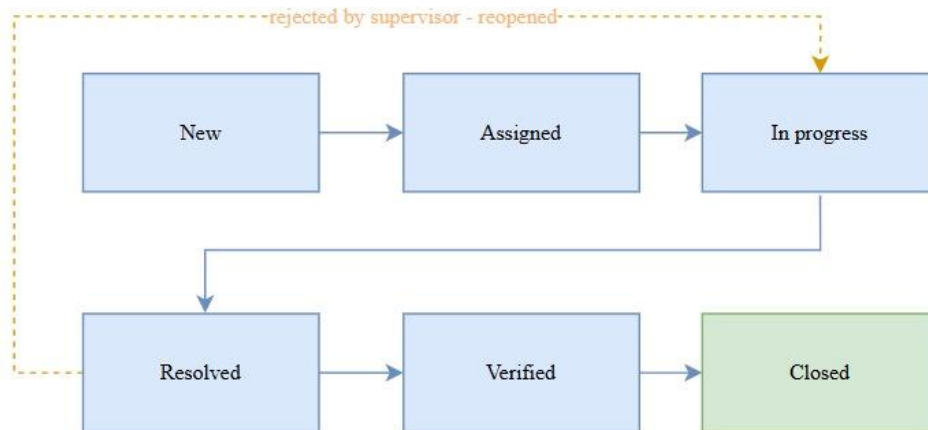


Fig. 2: Ticket Lifecycle

Two branches exit this path: RESOLVED returns to REOPENED if the supervisor rejects the repair, and NEW moves directly to AUTO_RESOLVED if the fault clears before anyone acts. Every transition records actor, timestamp and note. Staff cannot close their own tickets — the transition from RESOLVED to VERIFIED requires supervisor action.

Table 6: Ticket Categories, Priorities and SLAs

Category	Example Issues	Routed To	Priority	SLA
Electrical	Abnormal current, fan or light not working	Electrical Maintenance	High / Medium	2 h / 8 h
Cleaning	Board requires cleaning, cleanliness	Housekeeping	Low	24 h
Furniture	Missing, overturned or damaged furniture	Furniture Maintenance	Medium	8 h
Energy	Loads active in a vacant classroom	Electrical Supervisor	Medium	8 h

6.6 Classroom Infrastructure Health Score

A single 0-100 index per room, computed hourly, allowing a supervisor to rank many classrooms at a glance:

$$CIHS = 100 - (0.40 \times P_{\text{electrical}} + 0.20 \times P_{\text{furniture}} + 0.15 \times P_{\text{board}} + 0.25 \times P_{\text{maintenance}})$$

Scores of 85–100 are Normal, 60–84 Warning, and below 60 Critical. Weights are configurable and will be tuned with input from the institution's maintenance staff.

7. Privacy, Ethics and Safety

No raw video ever leaves the classroom — the edge device transmits only structured JSON observations, and frames are analysed in memory and discarded rather than stored. An evidence image is captured only at the moment a fault is confirmed. It is cropped to the relevant region, every detected person is blurred beyond recognition before the file is written, and the image is deleted after 30 days regardless of ticket status. The system has no facial recognition or biometric capability anywhere in its pipeline — occupancy is recorded as a count, never an identity — and no audio is captured at any point. Installation proceeds only after written permission from the Head of Department and institute administration, with signage displayed in the instrumented classroom to keep occupants informed.

Electrical safety. All mains-voltage work is confined to a purpose-built enclosed demonstration panel with a 6 A MCB, fuse protection and earthing, assembled and inspected under qualified supervision. No connection is made to classroom mains wiring at any point.

8. Hardware and Software Requirements

8.1 Hardware - Bill of Materials

Table 7: Hardware Bill of Materials

No.	Item	Qty	Purpose	Cost (Rs.)
1	Edge compute - Raspberry Pi 4 (2 GB), borrowed from the institute IoT laboratory	1	Edge AI processing	0
2	microSD card, 32 GB	1	Edge OS and storage	400
3	Camera - spare Android phone as a Wi-Fi IP camera	1	Classroom capture	0
4	ESP32 DevKit V1	1	IoT sensing node	400
5	PZEM-004T v3 energy module	1	V, I, P, PF, energy over UART	1,200
6	SCT-013-030 CT clamp	1	Independent fan-branch current	500
7	DHT11 sensor	1	Temperature and humidity	80
8	LDR module	1	Light-state cross-check	30
9	2-channel opto-isolated relay module	1	Demo-panel load switching	150
10	Demo panel - enclosure, 6 A MCB, fuse holder, terminal blocks, cable, earthing	1	Safe mains demonstration	1,200
11	Small 230 V table fan	1	Demonstration load	800
12	LED bulb with holder	2	Demonstration load	150
13	Jumper wires, perfboard, connectors, fasteners	-	Assembly	400
14	5 V USB supply for ESP32	1	Node power	150
Total				5,460

Contingency, if borrowed/existing items are unavailable: Raspberry Pi 4 with PSU and case Rs. 4,700; USB webcam Rs. 800; spare ESP32 Rs. 400 — worst case $\approx$ Rs. 11,400.

A Raspberry Pi 5 paired with a dedicated camera module was considered and set aside. At a 60-second sampling interval, a single YOLO-nano inference pass takes 2–4 seconds regardless of which board runs it, leaving the extra compute of a Pi 5 idle for the remainder of each cycle — the

bottleneck is the sampling design, not processor headroom. A mid-range phone camera also brings better optics and autofocus than the Pi's own camera module, and streaming over Wi-Fi decouples where the camera sits from where the compute sits, which matters more in a classroom retrofit than raw processing power does.

8.2 Software Stack

Table 8: Software Stack

Layer	Technology
Edge vision	Python, OpenCV, Ultralytics YOLO (nano), ONNX Runtime
IoT firmware	C++ on the Arduino framework for ESP32
Messaging	MQTT via Eclipse Mosquitto
Backend	Python, FastAPI
Database	PostgreSQL 16
Web client	React 18, Vite, TypeScript, Tailwind CSS, Recharts
PWA layer	vite-plugin-pwa (Workbox); IndexedDB offline queue
Notifications	Web Push (VAPID, RFC 8292) with pywebpush; Telegram Bot API as a redundant channel
Deployment	Docker Compose on a laboratory PC; Cloudflare Tunnel for HTTPS

Hosting cost: nil. The entire stack runs on one machine on the college LAN, removing dependency on campus internet connectivity during the viva.

Why a PWA rather than a native application. Once installed to the home screen, a PWA has its own icon, launches full-screen, works offline via its service worker and receives push notifications. One codebase serves both staff (phone, field use) and supervisor (desktop dashboard, calibration). Known limitation: Safari supports Web Push only on iOS 16.4+ and only when installed to the home screen — the Telegram channel covers this gap.

9. Expected Outcomes

1. A working single-classroom prototype performing automated periodic inspection.
2. Automatic generation of correctly classified, prioritised and routed maintenance tickets.
3. An installable web application serving supervisor and maintenance-staff roles with push notifications.
4. Measured detection accuracy and measured operational false-alarm rate, reported with variance across repeated trials rather than as single-run figures.
5. Demonstrated energy-wastage detection with quantified estimated wastage.
6. A complete digital audit trail from detection to verified closure.
7. Calibration tooling and a codebase that extends to further classrooms without retraining.

9.1 Performance Parameters

Table 9: Performance Parameters and Evaluation Targets

Parameter	Target	Measurement Method
Occupancy count error	MAE at most 2 seats	Against manual count, 100+ frames
Seat-level occupancy accuracy	At least 92 %	Per-ROI ground truth, held-out set
Board state agreement	At least 85 %	Versus two independent human annotators
Light ON/OFF detection	At least 98 %	Controlled switching trials
False ticket rate	At most 1 per classroom per day	7-day continuous unattended run
Missed fault rate	At most 5 %	Injected fault trials
Detection-to-ticket latency	At most 6 min from onset	Timestamped fault injection
Ticket-to-notification latency	At most 10 s	Server log versus device receipt
Edge inference time	At most 3 s per cycle	Instrumented timing
Fan rotation detection (E1)	At least 95 %	Controlled stop/start trials

These are design targets against which the prototype will be evaluated, not results already achieved — where a target is missed, the measured figure will be reported alongside an analysis of the likely cause rather than adjusted after the fact. No accuracy target is set for visible furniture-damage classification (E3): a dataset of damaged classroom furniture large enough to train a reliable classifier isn't practically collectible within this project's scope. Frames per second is deliberately excluded as a metric, since the system samples once every 30–60 seconds by design rather than aiming for high-rate capture.

Statistical reporting. Where feasible — occupancy accuracy, board-state agreement, false-ticket rate — each figure will be computed across multiple independent trials rather than a single run, and reported as mean, standard deviation and a 95% confidence interval, so the stability of the result is visible rather than assumed. Occupancy and board-state accuracy will additionally be reported as a confusion matrix against the held-out ground truth, in the same form used to evaluate the classification results in Section 9.

Computational cost. Edge inference time and duty cycle (≤ 3 s per cycle, under 7% duty cycle at a 60 s sampling interval) will be reported alongside detection accuracy, so accuracy figures are read against the resource budget they were achieved under rather than in isolation.

10. Applications and Social Relevance

The system reduces manual inspection workload, shortens the time between a fault occurring and being repaired, improves the availability of classroom electrical equipment, reduces energy wasted in unoccupied rooms, and creates a digital maintenance record that supports both accountability and data-driven preventive planning.

Beyond classrooms, the same architecture applies wherever infrastructure condition must be monitored across many similar rooms — laboratories, libraries, seminar and conference halls, training institutes, corporate offices and government buildings.

11. Conclusion and Future Scope

The proposed **Smart Classroom Infrastructure Monitoring and Maintenance System (SIMMS)** combines computer vision, IoT sensing, and digital maintenance management to monitor classroom infrastructure and identify abnormal conditions. The system uses occupancy and visual information along with electrical and sensor-based measurements to verify faults before generating maintenance tickets. Confirmed issues are classified, prioritised, assigned, and tracked through a digital workflow until supervisor-verified closure. The proposed approach aims to reduce false maintenance alerts, improve accountability, and maintain a complete digital record of classroom infrastructure issues.

The current work focuses on developing and evaluating a single-classroom prototype with core monitoring and maintenance functions. Future work will extend the system with vision-based fan rotation verification, furniture anomaly and damage detection, historical analysis for recurring faults, automated maintenance reports, and multi-classroom deployment. These extensions can further improve the system's ability to support preventive maintenance and large-scale classroom infrastructure management.

12. References

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